

Zomato Restaurant Analysis

1. Introduction

The restaurant industry generates vast amounts of customer and operational data every day. Converting this data into meaningful business insights enables restaurant owners and stakeholders to improve customer satisfaction, optimize operations, identify growth opportunities, and make informed strategic decisions.

This project presents an interactive Power BI dashboard developed using the Zomato Restaurant Dataset. The dashboard analyzes restaurant performance across multiple countries and cities by examining customer ratings, cuisine popularity, pricing, customer engagement, online delivery services and expansion opportunities. The report is divided into five analytical pages, each designed to answer a specific business problem while providing actionable insights.

2. Project Objectives

The primary objectives of this dashboard are:

- Analyze restaurant distribution across countries and cities.
- Evaluate customer satisfaction using restaurant ratings and votes.
- Study customer engagement and restaurant popularity.
- Identify the most popular and highest-rated cuisines.
- Compare restaurant performance across different price segments.
- Evaluate online delivery and table booking services.
- Identify high-potential cities for business expansion.
- Provide AI-inspired business recommendations for decision-making.

3. Dataset Overview

The dashboard is built using the Zomato Restaurant Dataset, which contains restaurant information from multiple countries.

Dataset Summary

Attribute	Value
Total Restaurants	9,551
Countries	15

Attribute	Value
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Cities	141
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Cuisine Categories	255
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Important Columns

- Restaurant Name
- Country
- City
- Cuisine
- Aggregate Rating
- Votes
- Average Cost for Two
- Price Range
- Online Delivery
- Table Booking

4. Tools & Technologies

Tool	Purpose
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Power BI Desktop	Dashboard Development
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Power Query	Data Cleaning
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DAX	KPI & Measure Creation
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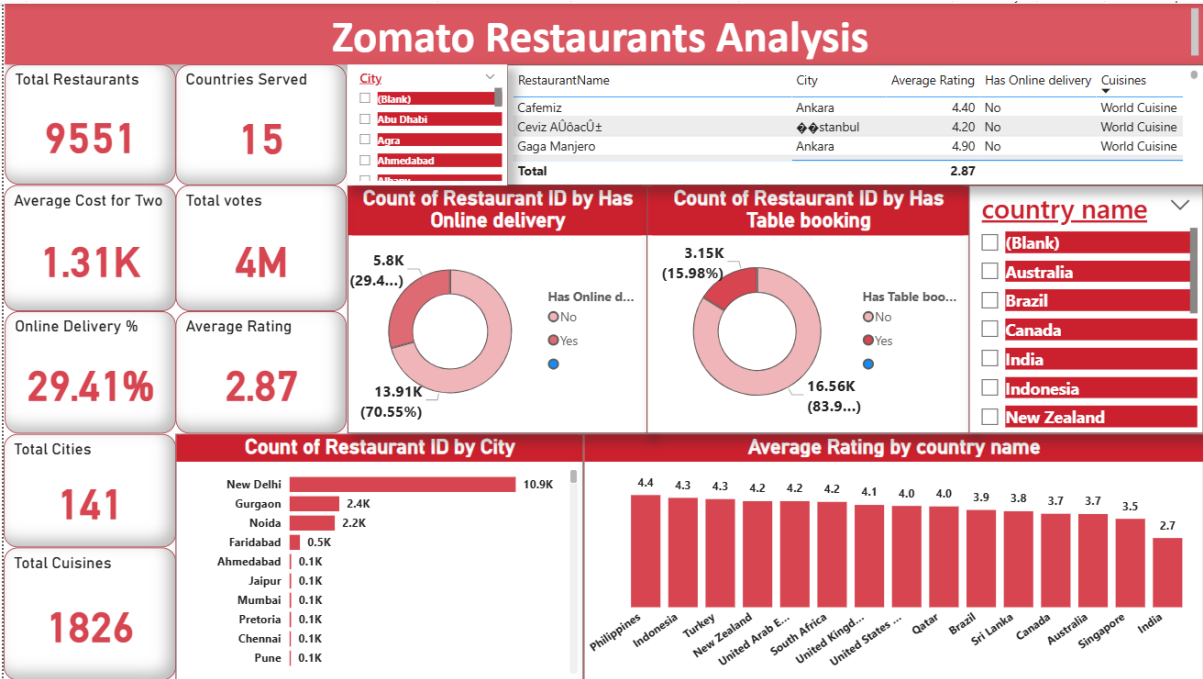
Microsoft Excel	Data Source
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GitHub	Project Repository
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5. Dashboard Overview

The dashboard consists of **five interactive pages**, each focusing on a different business perspective.

Executive Overview



Average Cost for Two

Represents the average dining cost for two people.

Business Value

Helps understand overall pricing strategy and customer affordability.

Total Votes

Displays the total customer votes received by all restaurants.

Business Value

Acts as an indicator of customer engagement and restaurant popularity.

Online Delivery %

Shows the percentage of restaurants providing online delivery services.

Insight

Only around **29%** of restaurants provide online delivery, indicating significant opportunities for expansion.

Average Rating

Represents overall customer satisfaction.

Insight

The average customer rating is **2.87**, suggesting there is room for service improvement.

Total Cities

Displays the number of cities covered.

Total Cuisines

Shows the diversity of cuisines available.

Visual 1

Online Delivery Availability (Donut Chart)

Purpose

Compares restaurants offering online delivery with those that do not.

Insight

The majority of restaurants still operate without online delivery, highlighting potential growth opportunities.

Visual 2

Table Booking Availability (Donut Chart)

Purpose

Shows restaurants offering table booking services.

Insight

Most restaurants do not provide table booking, indicating another area for service enhancement.

Visual 3

Top Cities by Restaurant Count

Purpose

Ranks cities based on the number of restaurants.

Insight

New Delhi dominates with the highest restaurant concentration, followed by Gurgaon and Noida.

Visual 4

Average Restaurant Rating by Country

Purpose

Compares customer ratings across countries.

Insight

Countries such as the Philippines and Indonesia exhibit higher customer satisfaction, whereas India records comparatively lower average ratings.

This page provides executives with a quick overview of business size, market presence, customer satisfaction, and operational services.

Customer Experience Analytics



This page focuses on customer satisfaction, engagement, restaurant popularity, and pricing.

Top Restaurants by Average Rating

Displays restaurants with the highest customer ratings.

Insight

Restaurants appearing at the top consistently deliver superior customer experiences.

Visual 2

Impact of Online Delivery on Customer Ratings

Compares average ratings between restaurants with and without online delivery.

Insight

Restaurants offering online delivery generally receive better customer ratings, suggesting greater customer convenience contributes to satisfaction.

Visual 3

Price Range vs Rating & Customer Engagement (Scatter Plot)

Analyzes the relationship between pricing, ratings, and customer votes.

Insight

Mid-to-premium restaurants tend to receive higher customer engagement and better ratings compared to budget restaurants.

Visual 4

Customer Ratings & Votes Across Price Ranges

Compares average ratings and customer votes across different price categories.

Insight

Price Range 3 receives the highest customer engagement, while Price Range 4 achieves the highest customer ratings.

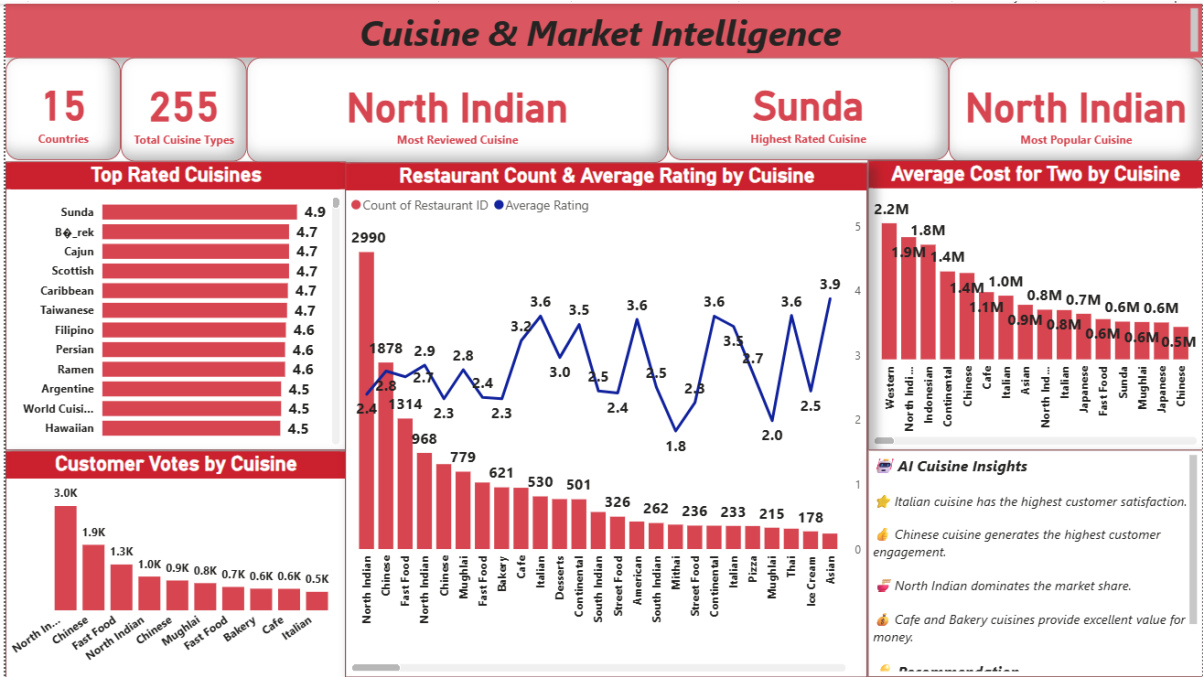
Business Questions Section

This section highlights the key analytical questions answered by the dashboard, helping users understand the purpose of each visualization.

Page Summary

The page demonstrates how pricing strategies and delivery services influence customer satisfaction and engagement.

Cuisine & Market Intelligence



Purpose

This page analyzes cuisine performance, popularity, customer engagement, and pricing trends.

KPI Cards

- Countries Served
- Cuisine Categories
- Most Reviewed Cuisine
- Highest Rated Cuisine
- Most Popular Cuisine

These KPIs identify cuisines that dominate customer preferences.

Visual 1

Top Rated Cuisines

Ranks cuisines according to average customer ratings.

Insight

Sunda cuisine receives the highest customer ratings, reflecting exceptional customer satisfaction.

Visual 2

Restaurant Count & Average Rating by Cuisine

Compares restaurant availability with customer ratings.

Insight

North Indian cuisine dominates the market, while several niche cuisines achieve higher ratings despite fewer restaurants.

Visual 3

Average Cost for Two by Cuisine

Analyzes pricing across different cuisines.

Insight

Western and North Indian cuisines are among the most expensive dining options.

Visual 4

Customer Votes by Cuisine

Shows customer engagement across cuisines.

Insight

North Indian cuisine receives the largest number of customer votes, indicating high popularity.

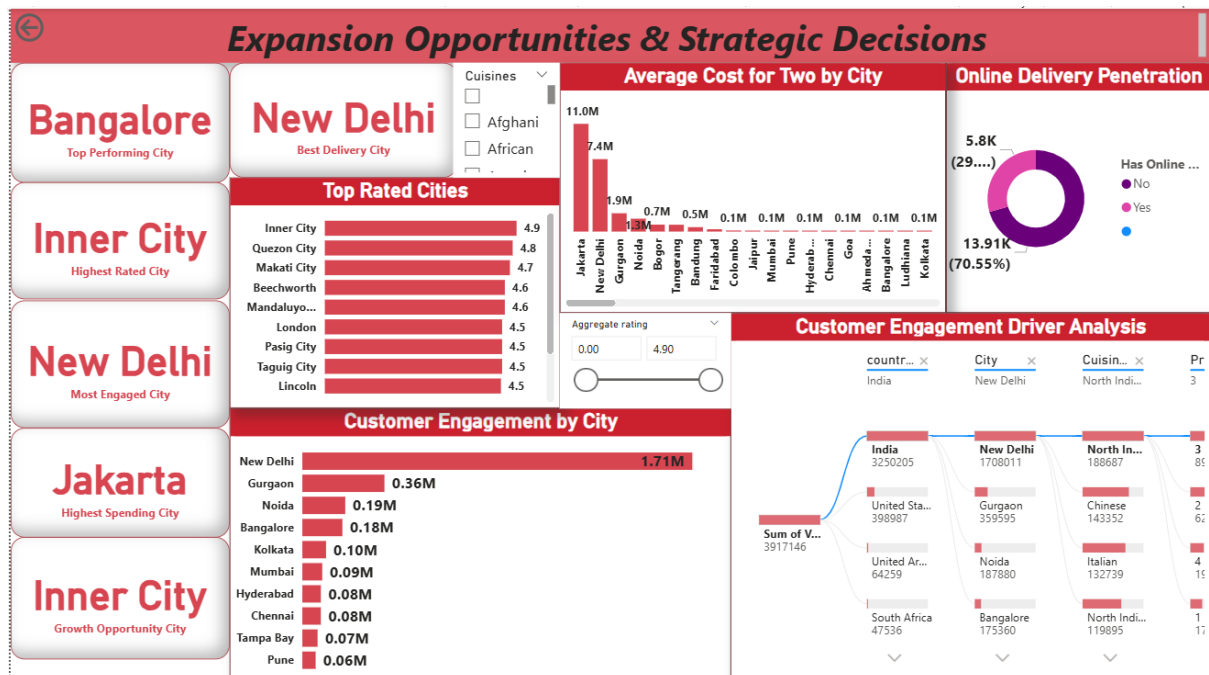
AI Cuisine Insights

Provides AI-generated business observations and strategic recommendations based on cuisine performance.

Page Summary

The page enables restaurant owners to identify high-performing cuisines and evaluate pricing strategies.

Expansion Opportunities & Strategic Decisions



Purpose

This page supports strategic decision-making by identifying high-performing cities and future expansion opportunities.

KPI Cards

- Top Performing City
- Best Delivery City
- Highest Rated City
- Most Engaged City
- Highest Spending City
- Growth Opportunity City

These KPIs summarize city-level business performance.

Visual 1

Average Cost for Two by City

Compares average dining costs across cities.

Insight

Jakarta and New Delhi represent premium restaurant markets.

Visual 2

Top Rated Cities

Ranks cities according to customer satisfaction.

Insight

Inner City receive the highest customer ratings.

Visual 3

Online Delivery Penetration

Displays the proportion of restaurants providing online delivery.

Insight

Online delivery remains underutilized, presenting opportunities for future expansion.

Visual 4

Customer Engagement by City

Ranks cities based on customer votes.

Insight

New Delhi generates the highest customer engagement.

Visual 5

Customer Engagement Driver Analysis (Decomposition Tree)

Breaks down customer engagement by country, city, cuisine, and price range.

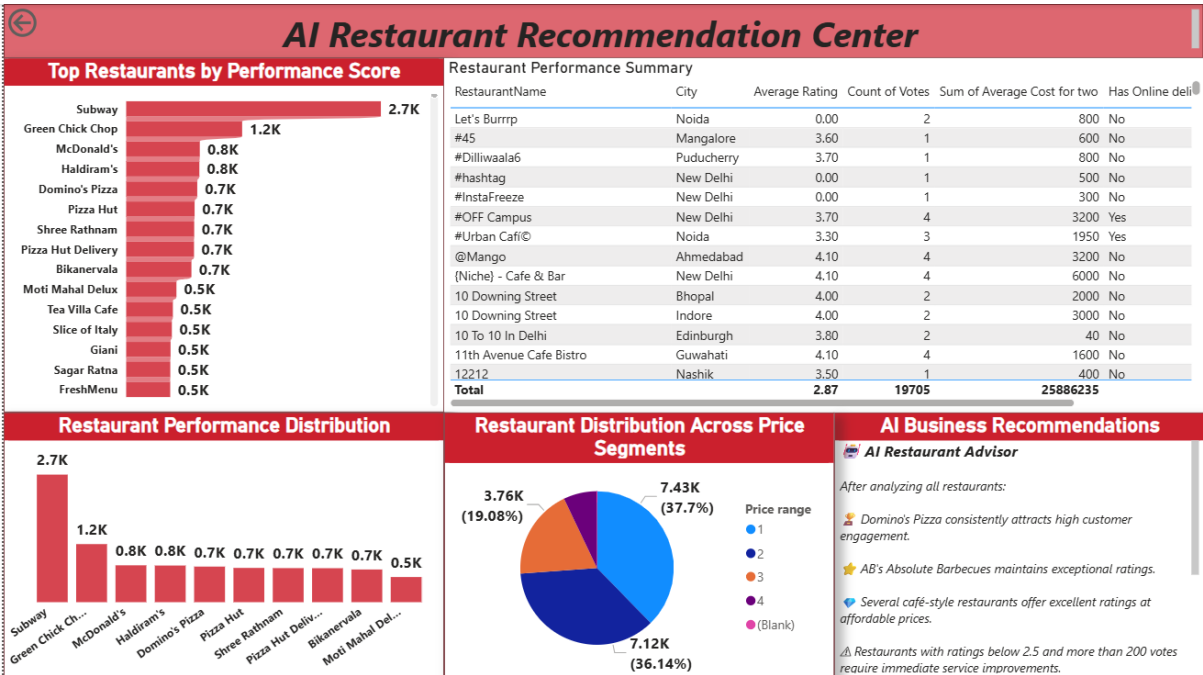
Business Value

Helps decision-makers identify factors contributing to higher customer engagement.

Page Summary

This page helps management identify investment opportunities and prioritize city-level expansion strategies.

AI Restaurant Recommendation Center



Purpose

The final page transforms analytical findings into actionable business recommendations.

Visual 1

Top Restaurants by Performance Score

Ranks restaurants using a custom performance score combining ratings, customer votes, affordability, delivery availability, and booking services.

Insight

Restaurants with the highest scores demonstrate superior overall performance.

Visual 2

Restaurant Performance Summary

Provides a detailed comparison of restaurant performance metrics.

Business Value

Supports restaurant benchmarking and operational evaluation.

Visual 3

Restaurant Distribution Across Price Segments

Shows the distribution of restaurants across different price ranges.

Insight

Most restaurants belong to the affordable and mid-range segments.

Visual 4

AI Business Recommendations

Provides intelligent recommendations derived from dashboard analysis.

Recommendations include:

- Promote high-performing restaurants.
- Improve low-rated restaurants.
- Expand online delivery services.
- Invest in premium markets.
- Focus marketing on highly rated but less popular restaurants.

6. Key Business Insights

- The dataset includes **9,551 restaurants** operating across **15 countries** and **141 cities**.
- New Delhi has the largest concentration of restaurants and the highest customer engagement.
- Only **29%** of restaurants offer online delivery, indicating a significant opportunity for digital expansion.
- North Indian cuisine dominates the market in terms of restaurant count and customer votes.
- Sunda cuisine records the highest customer ratings.
- Premium price segments generally achieve better customer ratings.
- AI-driven recommendations help identify high-performing restaurants and expansion opportunities.

7. Business Recommendations

- Expand online delivery services in high-demand cities.

- Promote highly rated but lesser-known restaurants through targeted marketing.
- Improve service quality for restaurants with low ratings.
- Increase investments in premium dining markets with strong customer engagement.
- Introduce table booking services in underserved markets.
- Leverage customer feedback to enhance restaurant performance and satisfaction.

8. Conclusion

The Zomato Restaurants Analysis Dashboard successfully transforms raw restaurant data into actionable business intelligence. Through interactive visualizations, KPI tracking, and AI-inspired recommendations, the dashboard provides valuable insights into restaurant performance, customer behaviour, cuisine trends, pricing strategies, and market expansion opportunities. It serves as a practical decision-support tool for business managers and stakeholders, enabling data-driven planning and strategic growth within the restaurant industry.
